DAFFODIL INTERNATIONAL UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FYDP (PHASE-I) EVALUATION REPORT

REPORTING PERIOD- SUMMER 2026

Project Identification:

I. Project Title	**An Explainable Deep Learning Framework for Deepfake Image Detection to Support Cybercrime Investigations**	
II. Group Members	1. Name: Md. Masum Al Mahamud Student ID: 0242310005101345 2. Name: Shinthiya Islam Ayshi Student ID: 0242310005101909	
III. Supervisor	Name: Dr. S. M. Aminul Haque Designation: Professor & Associate Head	
IV. Co-Supervisor	Name: Professor Dr. Md. Fokhray Hossain Designation: Dean & Professor	
V. Submission Date:	21 July 2026	
VI. Certificate:	"This is to certify that the final year design project work until Phase-I evaluation held on _____, titled as stated in Sec. I, executed by the students' group mentioned in Sec. II, have been found satisfactory and every section of this report is reflecting the same."	*(Signature of Supervisor & date)*

Project Insights

Thematic Area(s): ***[Just click the check box]***	Artificial Intelligence and Machine Learning	☑
	Deep Learning	☑
	Health Informatics	☐
	Cybersecurity	☑
	Software Engineering and Development	☐
	Blockchain Technology	☐
	Internet of Things (IoT)	☐
	Computer Networks	☐
	Computer Vision	☑
	Natural Language Processing (NLP)	☐

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	Robotics	☐
	Game Development	☐
	Cloud Computing	☐
	Image Processing	☑
	Others (please specify): Explainable AI (XAI) ☑	
Software packages, tools, and programming languages	Python 3; TensorFlow 2.20 and Keras; NumPy; Pandas; scikit-learn; Matplotlib; Pillow; Jupyter/Kaggle Notebook; Grad-CAM; Kaggle GPU (Tesla P100, 16 GB); Microsoft Word.	

CO Description for FYDP-Phase-I

CO	CO Descriptions	PO
CO4	Perform economic evaluation, cost estimation, and apply suitable project management procedures throughout the FYDP lifecycle in the context of developing the "Explainable Deep Learning Framework for Deepfake Image Detection" project.	**PO11**
CO6	Select and apply appropriate methodologies, resources, and contemporary engineering/IT tools for data preparation, deep learning, model evaluation, and explainable deepfake image detection.	**PO5**
CO7	Assess societal, safety, privacy, legal, ethical, and cultural issues and responsibilities associated with collecting public-figure images and using AI-generated or manipulated media in cybercrime investigations.	**PO6**
CO10	Operate effectively as individuals and as a collaborative project team during dataset preparation, model development, evaluation, documentation, and supervisor review.	**PO9**

1. Project Overview

1.1 Introduction

The aim with this project is to develop an explainable deep learning framework capable of classifying a digital image as Real or Fake and providing a visual explanation for the decision. The profiles aim to help cybercrime investigations related to A.I. generated photographs, impersonation, identity fraud, blackmail and false social media.

In Phase-I, we created an ethical collection protocol, prepared a balanced pilot dataset, created an end-to-end training and evaluation notebook, compared three transfer-learning architectures and ensemble, tested robustness under common social media distortions, and created Grad-CAM heatmaps for early explainability.

1.2 Background Study

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Modern diffusion and image-generation systems create realistic media that is difficult to detect using older GAN-focused detectors. Detector designs that account for diffusion-specific traces are required for deepfake detectors, according to Ricker et al. [1] Zheng and colleagues further demonstrated that detectors may learn semantic shortcuts rather than manipulation evidence which hinders cross-scene generalization.

Studies assessing performances on different images show that there are cases where a detector will perform worse than just a detector trained on their dataset [2]. Because investigators need more than a binary label; they need an indication of which image regions influenced the prediction, explainable approaches such as Grad-CAM have therefore become important [4], [7]. Low-resolution images, compression, benchmarking consistency and robustness, and interpretability are continuing research problems as recent work also notes [5], [6], [8].

1.3 Gap Analysis

Detectors that currently exist tend to report strong results within datasets and those that suffer from low resolution but are not robust toward unseen generators, disparate identities or scenes and social-media compression and noise. A lot of systems can also be black boxes, but in forensic and legal settings the explanations must be more transparent and cautious.

The current Phase-I experiment provides a working baseline. But it also indicates an important project-specific gap. Indeed, all 250 fake images in the pilot dataset came from one ChatGPT image generation source. Due to this, the present performance level does not show generalization to Gemini, DALL-E, FLUX, or other generators. The next phase should extend generator distribution, employ identity-disjoint and generator-aware evaluation, and check explanations under more realistic situations.

2. Objectives

The Phase-I objectives were to: (1) finalize the approved project scope and review recent deepfake-detection literature; (2) prepare ethical and traceable Bangladeshi public-figure image collection guidelines; (3) construct and audit a balanced pilot dataset containing 250 real and 250 fake images; (4) split the dataset before augmentation and build reproducible input pipelines; (5) train MobileNetV2, ResNet50V2, and Xception using transfer learning and controlled fine-tuning; (6) compare individual models with an equal-weight probability ensemble; (7) evaluate detection, robustness, and uncertainty using multiple metrics; and (8) generate Grad-CAM-based visual explanations and identify requirements for Phase-II.

3. Methodology/ Requirement Specification:

3.1 Research Design/ Prototype Design

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A design that is quantitative experimental was adopted. The system utilizes transfer learning to perform binary image classification (Real = 0, Fake = 1). All pictures were decoded as RGB, resized to 224 by 224 pixels, converted to float32 and passed through deterministic TensorFlow data pipelines. For reproducibility, we used deterministic operations when available, and fixed random seed 125.

Before augmentation, the pilot dataset was class-wise split into 70% train, 15% validation, and 15% test. A total of 348 images were kept for training samples. 76 images were kept in particular for validation and 76 in the testing dataset. Data augmentation was utilized only for the training split. Per epoch, every original training picture generated 1 additional augmented view using the following augmentations: horizontal flipping, small rotation, translation, zoom, mild contrast variation. This means 696 effective training views.

Using two-stage training, three ImageNet-pretrained backbones were made. Initially, the model's backbone was frozen while class-heads were trained. Second, the top 20% non-Batch-Normalization backbone layers were further fine-tuned using a slower learning rate. To prevent overfitting, several techniques were leveraged such as early stopping, model checkpoint, and learning rate reduction.

3.2 Data Collection/ Need Assessment

The data collection guideline defines a full target of 500 original real images from 50 Bangladeshi celebrities or public figures, with a planned balance of 25 male and 25 female identities and at least 10 accepted images per person [9]. Images should come only from public and traceable sources such as verified social-media pages, official websites, public press photographs, news portals, interviews, and clear public YouTube frames.

Quality screening prioritizes a clearly visible single face, sufficient facial resolution, varied lighting and pose, limited occlusion, and non-duplicate content. Private images, minors, heavily edited posters, strong beauty filters, ambiguous group photographs, and low-quality or heavily occluded images are excluded. Source URL, platform, collection date, image characteristics, label, fake-generation method, and split assignment are intended to be stored in a metadata sheet.

For the current Phase-I pilot, 250 accepted real images and 250 fake images generated from a single ChatGPT source were available for training and evaluation. The notebook verified image integrity, calculated SHA-256 hashes, removed exact duplicates, checked class balance, and recorded image dimensions. The expanded Phase-II dataset will add Gemini, FLUX, DALL-E, and other generators, with fake images generated separately within each split.

3.3 Analysis Techniques

Validation-only threshold selection was performed by searching thresholds from 0.05 to 0.95 and selecting the threshold that maximized F1-score, with balanced accuracy used as a secondary criterion. Individual models and an equal-weight probability ensemble were then evaluated on the untouched test set using accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC,

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average precision, Matthews correlation coefficient, confusion matrices, ROC curves, and precision-recall curves.

Bootstrap resampling was used to estimate 95% confidence intervals for the best candidate. Stability was tested under JPEG compression at quality 50, downsampling to 96 × 96 followed by restoration to 224 × 224, and Gaussian noise with sigma 8. Grad-CAM was applied to the strongest single model to highlight influential image regions. LIME and SHAP modules were prepared as optional extensions but were not executed in the present Phase-I run.

4. Progress Achieved:

4.1 Completed Tasks

Completed milestones include: proposal and title approval; literature review and gap analysis; preparation of the Bangladeshi public-figure dataset collection guideline; creation and auditing of a balanced 500-image pilot dataset; preprocessing, split, and augmentation pipeline implementation; two-stage training of three transfer-learning models; validation-based threshold optimization; ensemble construction; test, bootstrap, and stability evaluation; Grad-CAM explanation generation; and export of the best single deployment model, metadata, reports, and visual outputs.

The notebook now represents an end-to-end reproducible prototype: it discovers and audits the dataset, trains models, compares candidates, produces investigation-oriented outputs containing prediction, confidence, decision threshold and heatmap, and saves the selected model for later integration into a user interface.

4.2 Results Obtained

The quality audit retained 500 usable unique images—250 Real and 250 Fake—and found no exact duplicates. The split contained 174 Real and 174 Fake training images, 38 Real and 38 Fake validation images, and 38 Real and 38 Fake test images. With one augmented copy per training image, the effective training set contained 696 views per epoch.

The equal-weight ensemble was the best validation candidate. On the final test set it achieved 98.68% accuracy, 100.00% precision, 97.37% recall, 98.67% F1-score, 1.000 ROC-AUC, and 1.000 average precision at a threshold of 0.565. ResNet50V2 was the best single model for Grad-CAM and achieved 98.68% test accuracy with a 98.70% F1-score at a threshold of 0.375. MobileNetV2 and Xception each achieved 96.05% test accuracy.

The ensemble maintained 98.68% accuracy after JPEG quality-50 compression and after 96 × 96 downsampling followed by restoration. Gaussian noise reduced accuracy to 92.11% and F1-score to 91.43%, identifying noise robustness as a priority for improvement. The bootstrap 95% interval for test accuracy ranged from 96.05% to 100.00%. These findings are preliminary because the test set is small and all fake images were produced by one generator; they must not be interpreted as evidence of cross-generator generalization.

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5. Challenges Faced:

The main Phase-I challenges and planned responses are summarized below.

S.No.	Issues and Challenges	Strategies or Plans
1	Single-generator bias: the 250 fake images are from ChatGPT only.	Collect balanced fake subsets from Gemini, FLUX, DALL-E and additional generators; report per-generator and leave-one-generator-out results.
2	Small pilot dataset and possible identity overlap across splits.	Expand toward the planned 500 original real images, maintain full metadata, and adopt identity-disjoint train/validation/test partitions.
3	Risk of duplicate, edited, low-quality, or weakly traceable images.	Use SHA-256 and perceptual duplicate checks, quality/rejection logs, source URLs, and manual review before assigning labels.
4	Deep models and heatmaps can appear confident without providing forensic proof.	Use calibration and external testing, compare Grad-CAM with LIME/SHAP, document limitations, and retain human investigator review.

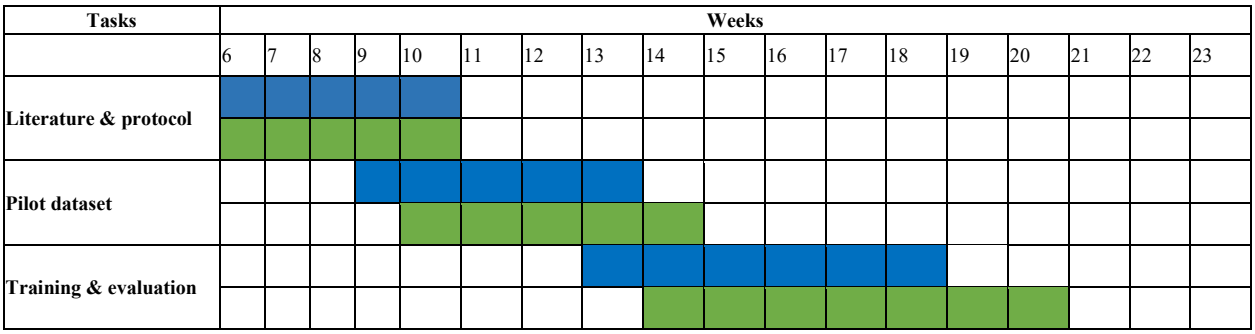
6. Next Steps:

The next phase will focus on dataset diversity, generalization, stronger explainability, and prototype integration.

S.No.	Next Task	Estimate completion time (MM-YY)
1	Collect, verify, label, and document multi-generator fake images from Gemini, FLUX, DALL-E and other available systems.	08-26
2	Create identity-disjoint and generator-aware splits, regenerate fakes within each split, and retrain all models.	09-26
3	Run external cross-dataset tests, leave-one-generator-out evaluation, calibration, and stronger distortion testing.	10-26
4	Compare Grad-CAM with LIME/SHAP, build an investigation-ready interface, and complete final documentation.	11-26

7. Updated Timeline:

The timeline was adjusted because the Phase-I model was completed using a pilot single-generator dataset; generator diversification and cross-dataset evaluation were moved to the next phase.



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Multi-generator extension																		

Estimated Work Period	
Actual Work Period	

8. Resources Utilized:

Resources used in Phase-I included a Kaggle Notebook environment with an NVIDIA Tesla P100 GPU (16 GB), TensorFlow 2.20/Keras, Python, NumPy, Pandas, scikit-learn, Matplotlib, Pillow, Jupyter Notebook, and Microsoft Word. ImageNet-pretrained weights were used for MobileNetV2, ResNet50V2, and Xception. Data resources consisted of 250 real public-figure images and 250 ChatGPT-generated fake images, together with the project collection guideline, proposal documentation, metadata plans, model checkpoints, CSV reports, and Grad-CAM visual outputs.

9. Project Management and Financial Analysis:

The project is managed through milestone-based work allocation between the two team members, periodic supervisor review, shared documentation, dataset logs, and versioned notebook outputs. Major milestones are data collection, quality control, model development, evaluation, explainability, interface integration, and final reporting. Risks are reviewed in relation to dataset bias, leakage, generator diversity, ethics, computational capacity, and false-positive impact.

Phase-I software and GPU costs were effectively minimized because open-source libraries and the Kaggle GPU environment were used. A provisional remaining budget of approximately BDT 8,000 is reserved for internet and data collection (Premium AI-BDT 2,500). This is an estimated planning figure and will be revised if paid generator access or additional computation becomes necessary.

10.Future Considerations:

Generalization is the most critical future concern. Visual artifacts can change rapidly owing to generator updates and new synthesis methods. Thus, the final dataset and evaluation protocol must cover multiple generator families and unseen generator testing. We need to control the identity leakage, shared prompts, repeated backgrounds, watermarks, filenames, or collection-source artifacts so that the detector learns image-forensic evidence rather than shortcuts.

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11.Conclusion:

Phase-I successfully produced a complete preliminary framework for explainable deepfake image detection. The team prepared a collection protocol, audited a balanced 500-image pilot dataset, trained and fine-tuned three transfer-learning models, selected thresholds using validation data, evaluated an ensemble, tested common distortions, and generated Grad-CAM explanations. The best ensemble achieved 98.68% accuracy on the small internal test set, while ResNet50V2 was selected as the explainable single model.

The results demonstrate technical feasibility but not final generalization. The current fake class is limited to 250 ChatGPT-generated images, and external cross-dataset testing has not yet been completed. Phase-II will therefore prioritize Gemini, FLUX, DALL-E and other sources, identity-disjoint and generator-aware evaluation, stronger robustness and calibration, comparison of explanation methods, and an investigation-ready prototype.

References

[1] J. Ricker, S. Damm, T. Holz, and A. Fischer, "Towards the Detection of Diffusion Model Deepfakes," in Proc. VISIGRAPP (VISAPP), 2024, pp. 446–457, doi: 10.5220/0012422000003660.

[2] C. Zheng, C. Lin, Z. Zhao, H. Wang, X. Guo, S. Liu, and C. Shen, "Breaking Semantic Artifacts for Generalized AI-Generated Image Detection," in Advances in Neural Information Processing Systems (NeurIPS), 2024.

[3] M. Brodarič, V. Štruc, and P. Peer, "Cross-Dataset Deepfake Detection: Evaluating the Generalization Capabilities of Modern Deepfake Models," 2024.

[4] J. J. Bird and A. Lotfi, "CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images," IEEE Access, 2024, doi: 10.1109/ACCESS.2024.3356122.

[5] K. Sharma, "Explainable AI-Generated Image Forensics: A Low-Resolution Perspective with Novel Artifact Taxonomy," in Proc. IEEE/CVF ICCV Workshops, 2025, pp. 1576–1585.

[6] N. Tasnim, K. Uddin, and K. M. Malik, "AI-Generated Image Detection: An Empirical Study and Future Research Directions," arXiv preprint arXiv:2511.02791, 2025.

[7] N. Bharati, K. C. P. Wong, S. Kouadri Mostéfaoui, D. Kbaier, and J. Collie, "Explainable Deepfake Detection: A Multi-Model Framework with Human-Interpretable Rationales for Legal Investigation Purposes," Machine Learning with Applications, vol. 23, Art. no. 100819, 2026, doi: 10.1016/j.mlwa.2025.100819.

[8] A. Mahara and N. Rishe, "Methods and Trends in Detecting AI-Generated Images: A Comprehensive Review," Computer Science Review, 2026.

[9] Project Team, "Bangladeshi Celebrity Image Dataset Collection Guidelines for Explainable Deepfake Image Detection," internal FYDP methodology document, Daffodil International University, 2026.

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Appendix

Appendix A: Phase-I Experimental Configuration and Preliminary Results

Item	Phase-I value
Usable pilot dataset	500 unique images: 250 Real and 250 Fake
Current fake-image source	ChatGPT only (250 images)
Split	Train 348; Validation 76; Test 76; balanced within every split
Training augmentation	One additional augmented view per training image; 696 effective views/epoch
Input size / batch size	224 × 224 RGB / batch size 16
Models	MobileNetV2, ResNet50V2, Xception, and equal-weight ensemble
Training method	Frozen classification head followed by controlled top-20% fine-tuning
Explainability	Grad-CAM on ResNet50V2; optional LIME/SHAP prepared but not run
External cross-dataset test	Not completed in Phase-I

Table A1: Final internal test performance

Candidate	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC
Ensemble	0.565	0.9868	1.0000	0.9737	0.9867	1.0000
ResNet50V2	0.375	0.9868	0.9744	1.0000	0.9870	0.9986
MobileNetV2	0.205	0.9605	0.9268	1.0000	0.9620	0.9993
Xception	0.505	0.9605	0.9487	0.9737	0.9610	0.9952

Note: These results are preliminary and are based on a small internal test set with a single fake-image generator.

Table A2: Ensemble stability under social-media-style distortions

Condition	Accuracy	F1	Average Precision
Original	0.9868	0.9867	1.0000
JPEG quality 50	0.9868	0.9867	0.9993
Downsample 96 then restore	0.9868	0.9867	0.9993
Gaussian noise sigma 8	0.9211	0.9143	0.9979

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Appendix B: Notebook-Generated Figures

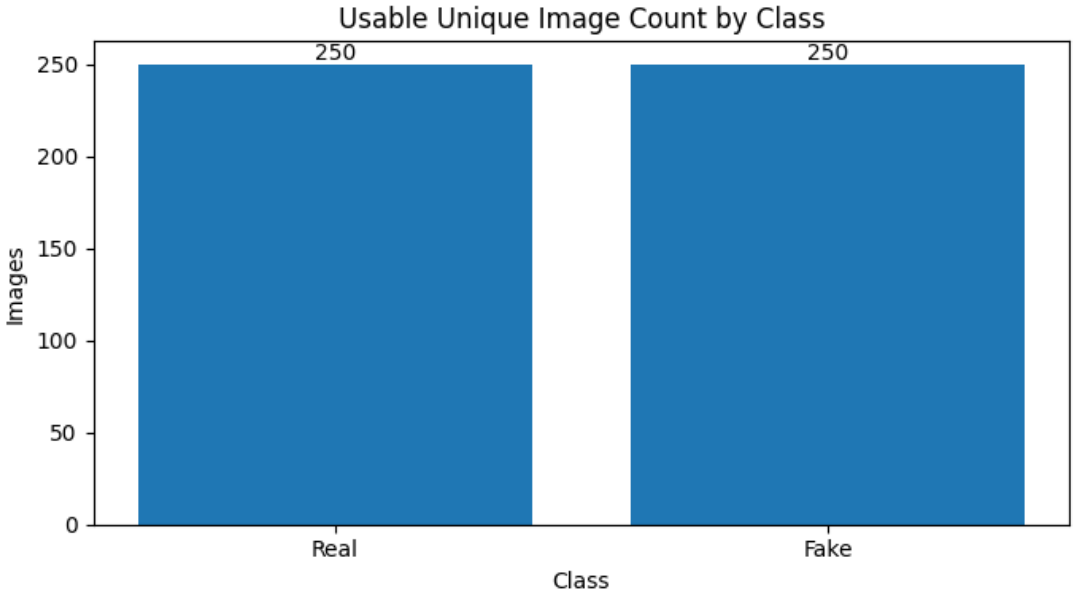


Figure B1: Balanced class distribution after integrity and duplicate checking.

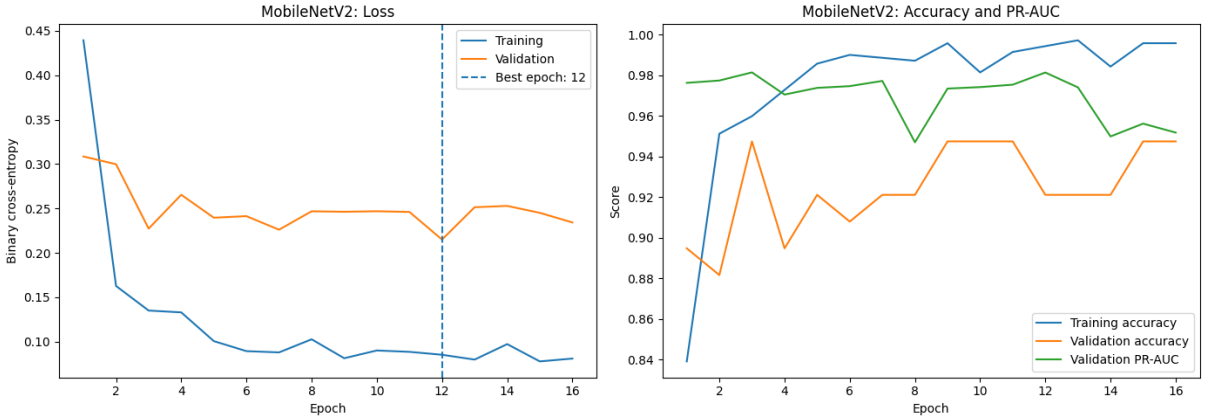


Figure B2: MobileNetV2 training and validation curves.

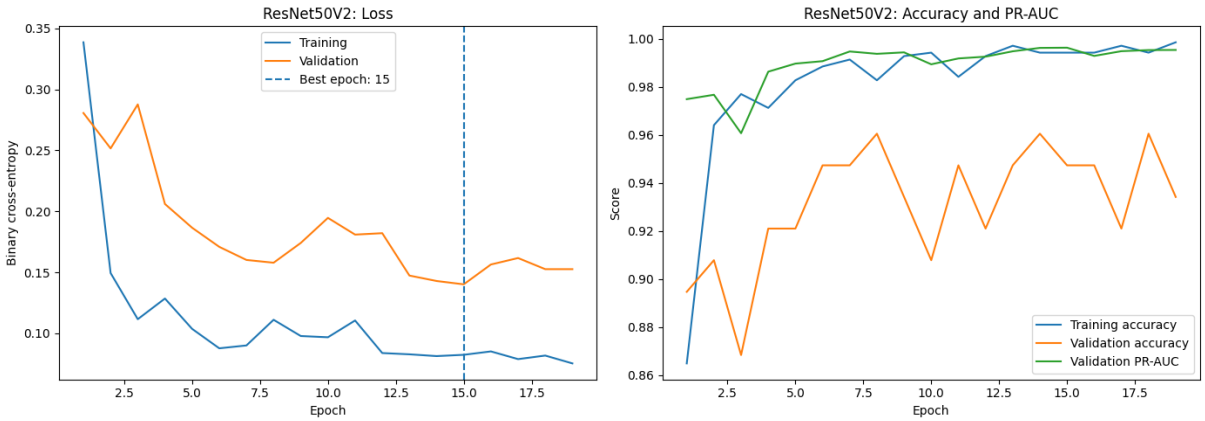


Figure B3: ResNet50V2 training and validation curves.

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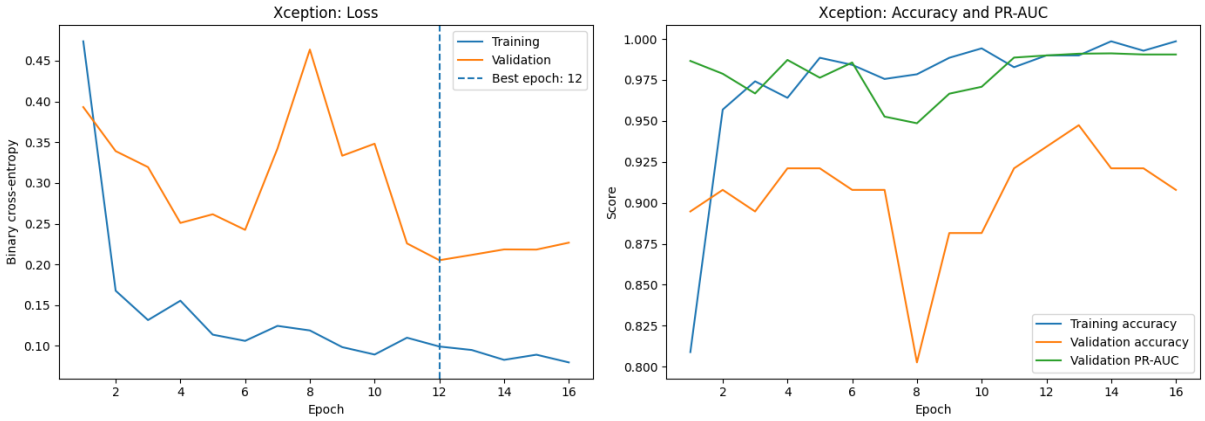


Figure B4: Xception training and validation curves.

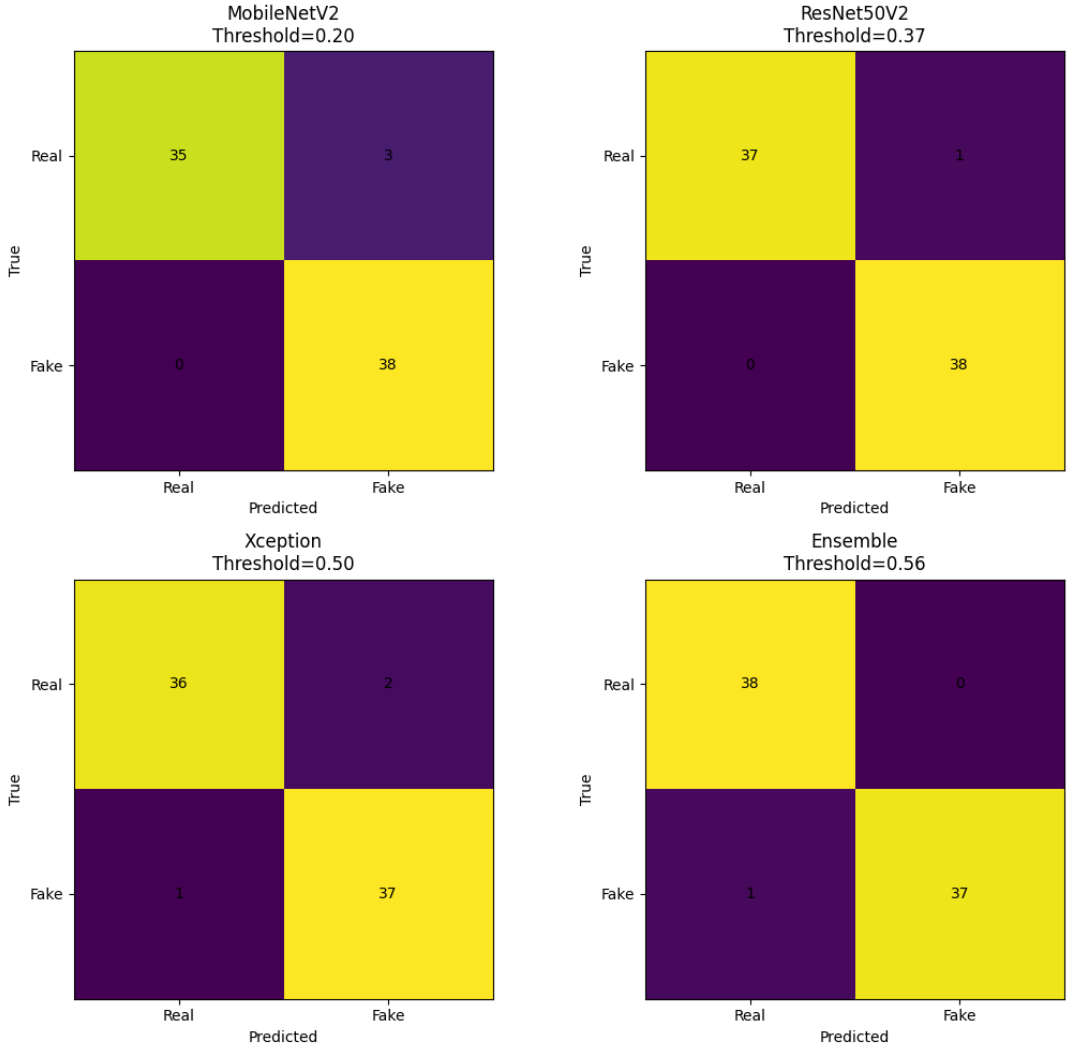


Figure B5: Test confusion matrices for the three models and the ensemble.

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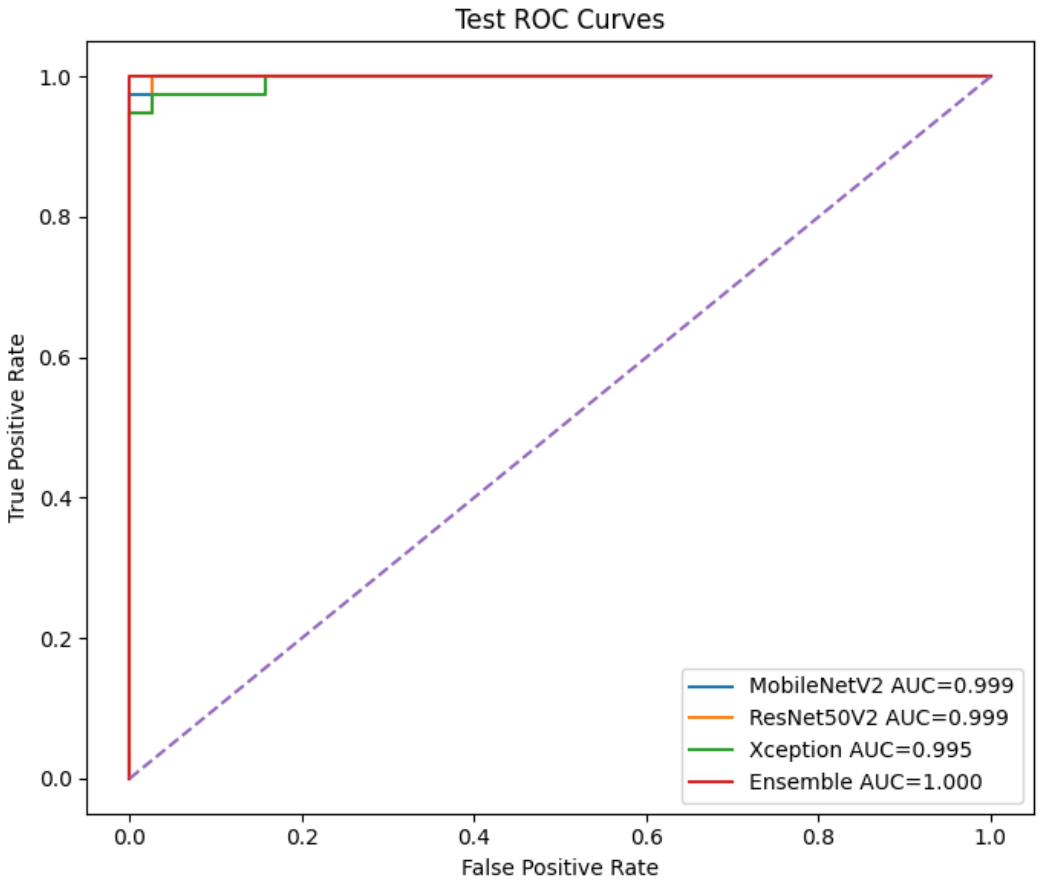


Figure B6: Test ROC curves.

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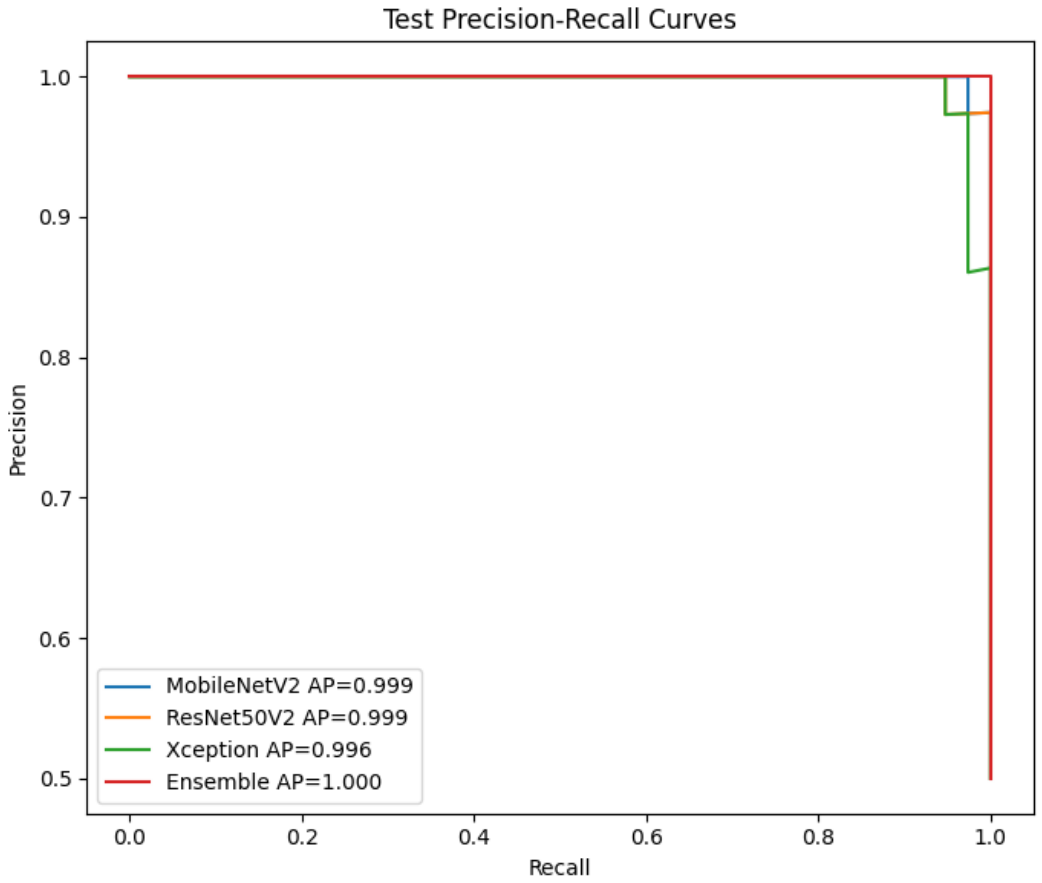


Figure B7: Test precision-recall curves.

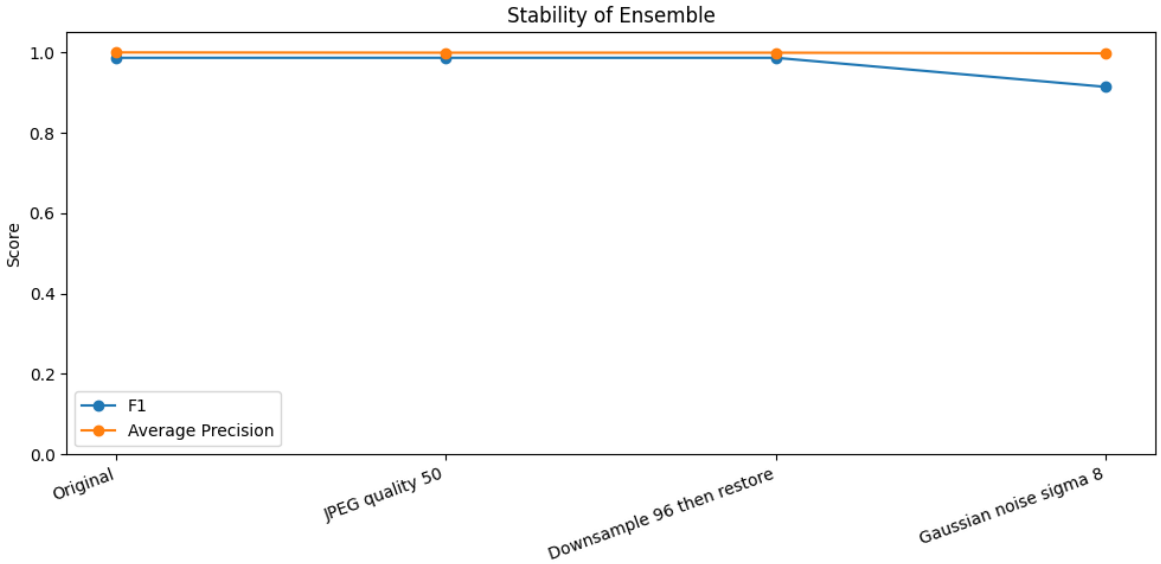


Figure B8: Stability of the ensemble under compression, resizing, and noise.

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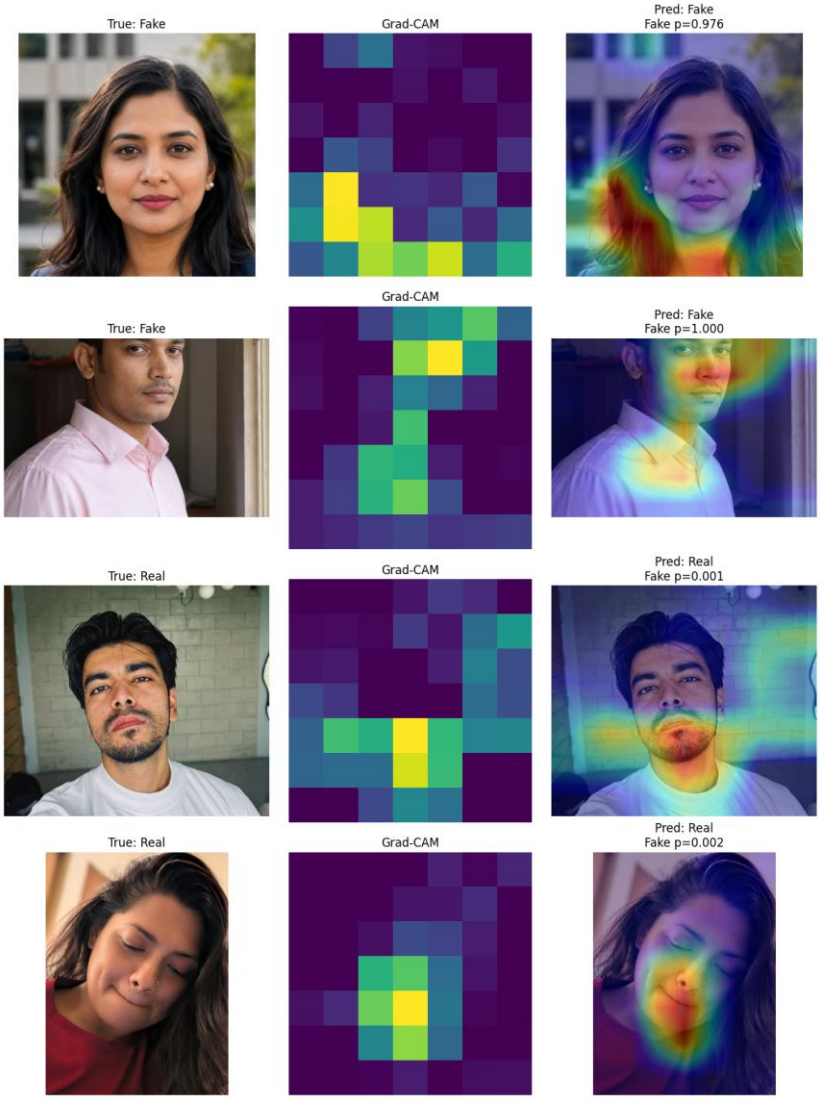


Figure B9: Grad-CAM examples from the untouched test set.

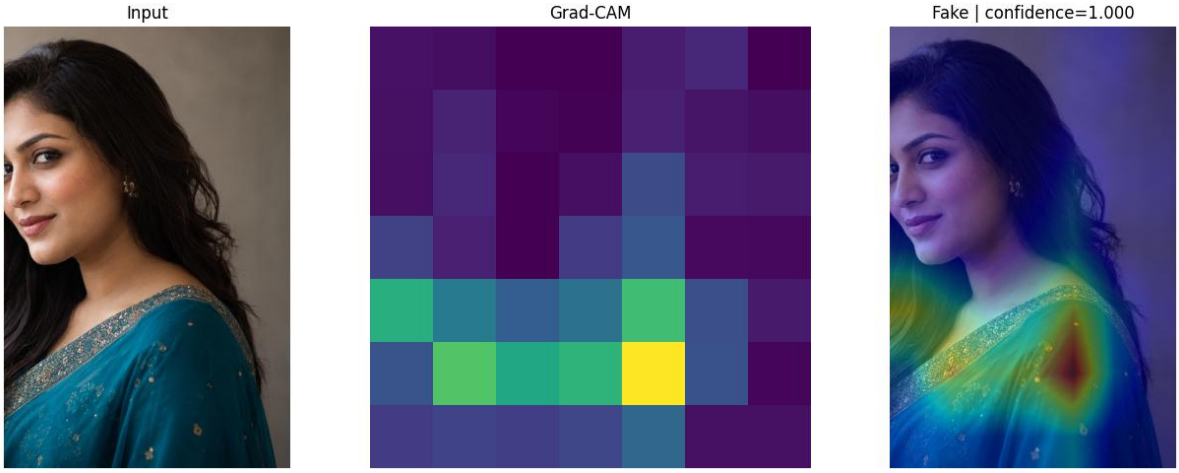


Figure B10: Investigation-ready single-image output showing the input, Grad-CAM map, and overlay.

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Appendix C: Planned Multi-Generator Dataset Extension

Generator/source	Phase-I status	Phase-II action	Purpose
ChatGPT	250 fake images used	Retain as one balanced generator subset	Baseline and within-source comparison
Gemini	Not yet included	Collect a balanced fake subset with recorded model/version and prompts	Cross-generator generalization
FLUX	Not yet included	Collect a balanced subset covering portraits and realistic social-media styles	Diffusion-family diversity
DALL-E	Not yet included	Collect a balanced subset with source metadata	Cross-provider evaluation
Other generators	Not yet included	Add available open and commercial generators where ethically and technically feasible	Leave-one-generator-out testing

The expanded evaluation will report overall metrics, per-generator metrics, identity-disjoint performance, and leave-one-generator-out performance. Generator names, model versions, prompts or generation settings, dates, and post-processing operations will be recorded in metadata.

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FINAL YEAR DESIGN PROJECT
PHASE-I PROGRESS REPORT

This report, in the form of a template, has been specifically designed for BSc. students working on their Final Year Design Project (FYDP) at Computer Science and Engineering Department, Daffodil International University (DIU).

Every group of students is required to do the following:

1. Complete all the sections of this template
2. Get it certified by the assigned supervisor before one week of Phase-I evaluation presentations
3. Submit 01 photocopy to each of the following, on or before the day of Phase-I presentations:
    a. Supervisor
    b. Internal Evaluator
4. Submit original copy to FYDP committee on the day of Phase-I presentations.

Note:

1. Use English
2. There should be NO grammatical or spelling mistakes
3. Submission after due date will not be accepted
4. For more information, contact your Supervisor

Template prepared by:	**Template approved by:**
FYDP Committee **Dept. of CSE, DIU**	**Dr. Sheak Rashed Haider Noori** **Professor and Head, Dept. of CSE, DIU**

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